

SkyStation Operations HQ

Complete Project Handoff and Current-State Context

A source-derived briefing for a new engineer, designer, operator, or AI agent taking over the project.

Document status	Current repository state
Generated	2026-09-19 18:06 UTC
Repository	akshobhyaa-karyachakra/SkyStation-Operations-HQ
Branch	main
HEAD	219d1ec (HEAD -> main, origin/main) docs: separate portal polish from wallboard redesign
Latest UI release	0d27e67
Documentation baseline	728d74f
Latest documentation correction	219d1ec
Test baseline	 [94%] [100%] 76 passed in 11.99s

Read this first: The repository contains a working synthetic portal preview, a separate six-screen Public Wallboard, protected adapter/server foundations, extensive source contracts, and ongoing uncommitted source/role-framework work. The static UI is published; production runtime integration and browser screenshot QA are not complete.

1. Executive summary

SkyStation Operations HQ is Skylark's shared operational review surface. It is intended to bring activity execution, inventory readiness, crew context, analytics, billing evidence, customer MIS reporting, and a public-safe display into one coherent workspace.

The current implementation is a single-file frontend preview plus Python server-side adapters and snapshot contracts. The frontend is deliberately synthetic and authentication-free in the preview. The production shape is protected server adapters → validated snapshots → public-safe or manager-authorized projections → browser surfaces.

The most important boundary is between the Public Wallboard and internal Manager Vault/deep-dive data. The wallboard is designed for a TV: six fixed screens, no scrolling, no mouse dependency, public-safe aggregates, automatic cycling, and explicit source/freshness states.

Current maturity: UI and source-contract foundations are substantial; production hosting, runtime secrets, OAuth, scheduler, live public projection, Manager Vault authentication, and real browser visual QA remain unfinished.

Current implementation status

Area	Status	Evidence / next action
Portal shell	Implemented preview	rebuild-preview.html; route buttons and embedded modules exist.
Public Wallboard	Implemented static preview	public-wallboard.html; six screens, controls, public-safety scan passed.
Server runtime	Implemented foundation	scripts/portal_server.py; protected snapshot routes and public-status route.
Source adapters	Broadly implemented	sync scripts for activity, flight, site, QA, reporting, inventory, incidents, billing, crew, roles, workflow.
Production integration	Not complete	Configure HTTPS runtime, secrets, scheduler, OAuth, storage, freshness window.
Manager Vault	Prototype/partial	Protected crew-portal routes exist; production auth and full UX are unfinished.
Browser visual QA	Pending	Browser harness was unavailable; source/runtime checks are not screenshot QA.
Role framework	Working-tree implementation	sync_role_framework.py and test_role_framework.py are uncommitted and must be reviewed before commit.

2. Product context and user intent

The product is not a generic SaaS dashboard. The intended experience is a Skylark Mission Control surface: precise, field-tested, practical, and useful in operational reviews. The screen should answer what is happening, why it matters, what evidence supports it, and what needs attention.

Sai's central design constraint is that the Public Dashboard is a TV wallboard, not a web application. Every screen must fit completely inside one 16:9 viewport, with no vertical or horizontal scrolling and no hover-only or click-only information.

The current internal portal and the Public Dashboard are now separate design tracks. The internal portal structure is frozen for future polish-only work. The Public Dashboard remains eligible for broader redesign.

User groups

- Internal operations teams: review execution, delivery, readiness, crew presence, and exceptions.
- Managers: inspect protected evidence, owners, blockers, crew detail, role framework, billing evidence, and one-to-one records.
- Customer-facing teams: prepare approved customer-safe MIS and delivery views.
- Public display viewers: see aggregate operational confidence and current operating signals without internal detail.

Explicit non-goals

- The browser must not call Monday directly.
- Public surfaces must not expose names, board/item/People IDs, billing values, serial numbers, credentials, internal notes, or unresolved root-cause narratives.
- The system must not infer utilization, workload, capacity gaps, or availability from missing records.
- Planning records do not prove execution, delivery, or report completion.
- Manager Vault production authentication is not complete in the current preview.

3. Thread decision history

The following decisions were made explicitly in the project thread and should be treated as product context, not incidental chat commentary.

Date	Decision
2026-09-17	Build a customer-safe Public Dashboard as a separate multi-screen TV wallboard.
2026-09-17	All relevant information for a wallboard topic must fit on the screen; no scrolling or mouse dependency.
2026-09-17	Add a dedicated Crew and Resource Pulse screen.
2026-09-17	Use Crew Repository Availability as the only public daily-presence source.
2026-09-17	Blank availability means Awaiting confirmation, never Available.
2026-09-17	Exclude Terminated and Resigned from active totals.
2026-09-17	Keep individual crew, workload, allocation, utilization, and review detail out of public display.
2026-09-17	Use a six-screen narrative: health → delivery flow → crew presence → geography → reporting → fleet readiness.
2026-09-17	Public Wallboard needs a broader redesign because the first designs had excessive blank space and insufficient data density.
2026-09-17	Fix portal icon consistency, Wallboard return navigation, and Analytics network first-load initialization.
2026-09-19	Use a living document system: PRD, logic index, IA, design, architecture, API, security, ADRs, QA, runbook, and decision log.
2026-09-19	Internal portal tabs remain structurally frozen for polish-only work; Public Dashboard stays a separate redesign track.

4. Repository and source-of-truth map

The repository is the implementation source. The existing business-logic Google Doc is intended to remain the canonical operations-rules source, but its URL and owner have not yet been recorded in the repository. Do not invent or guess that URL.

Path	Role
PRODUCT.md	Product purpose, users, brand personality, design principles, accessibility.
rebuild-preview.html	Main single-file internal portal preview; shell, routes, embedded styles, data, charts, crew, billing, analytics, MIS, and vault prototypes.
public-wallboard.html	Separate six-screen fixed TV wallboard.
index.html	Redirect-only entry point.
scripts/portal_server.py	Protected HTTP runtime, snapshot routes, auth checks, stale/invalid handling, crew-portal store, public status.
scripts/sync_*.py	Server-side source adapters and snapshot builders.

Path	Role
scripts/portal_metrics.py	Derived metrics builder from loaded snapshots.
tests/	Adapter, contract, metrics, server, workflow, billing, inventory, incident, crew, and role tests.
docs/data-contract.md	Normalized entities, source fields, relation rules, freshness, public projection.
docs/monday-source-map.md	Source board and column mapping.
docs/frontend-design-plan.md	Mission Control visual system, information hierarchy, responsive/motion plan.
docs/documentation-roadmap.md	Documentation governance and update cadence.
docs/business-logic-index.md	Index from the missing-link Google Doc to implementation.

5. Current portal implementation from code

File size: rebuild-preview.html is 390,431 bytes and 171 lines. It is a single-file frontend preview with large embedded CSS and JavaScript sections. It is not a componentized React application.

The portal's first-load route is Activity Tracking. The navigation uses client-side buttons and toggles the visibility of route sections rather than a server router.

Route key	Visible module	Current behavior
activity	Activity Tracking	Default visible route; execution and attention review.
inventory	Inventory	Fleet deployment and hardware inventory tabs; visual catalogue and register.
crew	Crew Management	Crew overview, role/band management, deep dive, responsibilities, performance, one-to-ones.
analytics	Analytics	Daily/weekly/monthly tabs, planned-vs-complete, heatmap, queues, workflow network.
billing	Billing	Preview billing analytics and filtering with source-state handling.
mis	Customer MIS Export	Customer-safe export preview route.
manager-vault	Manager Vault	Design/prototype boundary; production authentication incomplete.
public-wallboard	Public Wallboard	Full-page navigation to separate public-wallboard.html.

Portal modules visible in source

- Activity Tracking: planned flights, executed flights, blocked flights, late reports, attention panels, pipeline, trends, source context, and drill-down handlers.
- Inventory: fleet deployment, hardware inventory, asset gallery, paired/related visual cards, registers, and readiness context.
- Crew Management: aggregate crew views plus protected-style role, responsibility, performance, one-to-one, and deep-dive surfaces.
- Analytics: period controls, trend chart, heatmap, queue rows, handoff workflow, delay/weather/source signals, and a 3D workflow constellation.
- Billing: period/customer filters, billing rows, decision/source states, and preview-only behavior.
- MIS: customer-safe report/export preview boundary.
- Manager Vault: navigation and prototype route; production auth and authorization are not complete.

Embedded design systems

The source contains multiple embedded style blocks, including shared design system, brand mark asset, activity responsive grid, attention layout, portal typography/compact navigation, analytics consistency, workflow redesign, and final workflow layout. Before changing a visual rule, search all style blocks and the corresponding JavaScript initializer rather than editing only the first CSS declaration.

6. Public Wallboard implementation from code

File size: public-wallboard.html is 34,270 bytes and 30 lines. It contains six sections with data-screen values 0–5.

The page is a fixed viewport display. It uses CSS overflow hidden, a shared top bar, a left public-display rail, a stage containing one active screen at a time, and a footer with controls and progress.

The wallboard was intentionally separated from the internal portal because its public data boundary, TV interaction model, screen cycle, and design direction are different.

Screen	Purpose	Primary visual/data
01 Operations Pulse	Overall operational position	Health signal, sparkline, active sites, completed flights, attention, reports, execution rate, interpretation.
02 Delivery Flow	End-to-end handoff	Mapped → scheduled → flown → processed → reported → delivered, conversion tracks, evidence queue, delivery state.
03 Crew and Resource Pulse	Daily verified presence	Team/status matrix, active totals, presence mix, source quality, no inferred capacity.
04 Geographic Operations	Where work is active	Regional operating grid, aggregate site/work signals, attention by region.
05 Reporting Outcomes	What has been delivered	Planned/submitted trend, reporting cadence, pipeline, ageing/readiness signals.
06 Fleet and Readiness	Operational readiness	Readiness ring/constellation, categories, maintenance/attention, interpretation.

Controls and motion

- Automatic cycling is approximately 22 seconds per page.
- Space pauses/resumes; Left/Right changes screen; R restarts the cycle.
- The page pauses when the browser tab becomes hidden.
- Reduced-motion mode suppresses decorative animation while preserving final states.
- Entry motion animates panels, bars, sparks, flow nodes, reporting bars, and readiness visuals.
- The main portal return path is `rebuild-preview.html#activity`.
- The favicon and top-left mark use `assets/favicon.svg?v=2`.

7. Data model and source contracts

The data model is source-backed and relation-aware. Monday display names are not join keys. Every normalized source record carries provider/source identity, source update time, sync time, and a data state.

Important: The static wallboard currently contains public-safe preview values. The protected runtime and adapters are the intended path for live source-backed projections; do not silently treat the static values as live production data.

Entity	Canonical source / rule
crew_member	Crew Repository board 5030902067; Availability is public presence source; historical statuses excluded.
role_framework	Role Framework board 5031376976 and generated competency subitems; explicit parent/subitem IDs; weights must total exactly 100% for valid role records.

Entity	Canonical source / rule
planned_activity	Activity Repository 5027240228; planning/reference only; does not prove execution or delivery.
flight	Flight Operations 5027240883; execution stage.
site_activity	Site Activities 5028018276; execution/site stage.
processing_qa	Processing and QA 5027256991; processing stage.
report_submission	Report Submission 5027265596; reporting stage.
work_tracker	Work Tracker 5029561760; internal tracking; cannot prove customer delivery.
inventory_asset	Inventory 5028042389; asset identity/readiness/relations.
incident	Incident Logs 5030309792; incident state/evidence; no inferred RCA/closure.
billing	Customer Repository billing projection; protected; financial values excluded from public surface.

Freshness and failure states

- current: latest valid snapshot within configured freshness window.
- stale: last valid snapshot exists but is older than the window.
- unavailable: no valid snapshot is available.
- needs_review: snapshot exists but a contract invariant failed.
- Missing records are not converted to zero totals presented as real operational values.

8. Runtime and API details

The server is implemented in scripts/portal_server.py using Python's HTTP server machinery. It loads snapshot JSON from the repository data location or configured runtime paths, checks authorization, determines snapshot state, and emits JSON responses.

The server has PORTAL_API_TOKEN, optional loopback-only PORTAL_DEV_ALLOW_LOCAL=1, PORTAL_SNAPSHOT_MAX_AGE_SECONDS with a default of 86400 seconds, crew portal store/file paths, and a maximum request body size setting. Values must be supplied through runtime configuration, never committed.

Endpoint	Access / purpose
/api/public-status	Public; source states, schema versions, updated timestamps, overall state.
/api/crew	Protected Crew Repository snapshot.
/api/activity	Protected Activity Repository snapshot.
/api/flight	Protected Flight Operations snapshot.
/api/site-activities	Protected Site Activities snapshot.
/api/processing-qa	Protected Processing and QA snapshot.
/api/report-submission	Protected Report Submission snapshot.
/api/work-tracker	Protected Work Tracker snapshot.
/api/inventory	Protected Inventory snapshot.
/api/incidents	Protected Incident Logs snapshot.
/api/workflow-network	Protected workflow network snapshot.
/api/billing	Protected billing snapshot.
/api/role-framework	Protected role framework snapshot.
/api/metrics	Protected derived metrics from available snapshots.

Endpoint	Access / purpose
/api/crew-portal	Protected Manager Vault store.
/api/crew-portal/*	Protected role lenses, performance, responsibilities, requests, one-to-ones.

Response behavior

- Protected route without authorization: 401 unauthorized.
- Snapshot missing: 503 with data_state unavailable.
- Snapshot invalid: 503 with data_state needs_review.
- Valid/current or stale snapshot: 200, body plus data_state and snapshot_updated_at.
- Public status intentionally omits records and credentials.

9. Security and privacy boundary

The security design is projection-first. Hiding fields in the UI is insufficient; protected data must be filtered before it reaches the public browser surface.

Never preserve: credentials, API keys, OAuth secrets, passwords, tokens, connection strings, or raw sensitive values. If such a value is encountered in a document, log, or artifact, replace it with [REDACTED].

Publicly allowed when approved	Always protected/excluded
Team/region aggregates	Names and individual crew records
Approved counts/rates	Monday board/item/People/relation IDs
Source state and freshness	Billing/financial values
Customer-safe delivery/readiness outcomes	Asset serial numbers/replacement costs
Synthetic/awaiting-source labels	Internal notes, incident narratives, unresolved RCA

Production requirements still open

- Deploy portal_server.py behind production HTTPS.
- Configure PORTAL_API_TOKEN and MONDAY_API_TOKEN in a secret manager.
- Configure snapshot storage, retention, freshness window, timezone, sync cadence, and stale-read policy.
- Configure Google OAuth and the approved manager group/allowlist.
- Run unauthorized/authorized/missing/invalid/stale requests against deployed HTTPS.
- Verify browser receives only the authorized projection.

10. Tests, verification, and release state

The repository test suite currently reports 76 passed. The tests cover adapter contracts and validation for activity repository, flight operations, site activities, processing/QA, report submission, work tracker, inventory, incidents, workflow network, billing, portal metrics/server behavior, crew repository, and role framework files present in the working tree.

The latest verified static UI checks include JavaScript parsing for both HTML documents, six wallboard screens, favicon/back-link wiring, Analytics post-layout initialization, public-safety checks, and git diff --check.

The browser harness was unavailable during prior verification. Therefore screenshot-based claims about typography, clipping, canvas rendering, exact TV-distance readability, and cross-viewport appearance remain unverified.

Check	Current state
Repository tests	76 passed at handoff generation.
JavaScript syntax	Passed for portal and wallboard inline scripts.
Public wallboard contract	Six screens; fixed/no-scroll source contract passed.
Public safety	Passed against known prohibited content checks.
Deployment	GitHub Pages workflows successful through documentation commits.
Browser screenshot QA	Pending; harness unavailable.
Production runtime QA	Pending; external HTTPS/auth/scheduler/secrets not configured.

11. Git history and current working tree

The repository is on main. The recent history is intentionally useful for understanding the build sequence: portal foundation and adapters, analytics workflow, billing, role framework, wallboard, documentation roadmap, documentation baseline, and portal/wallboard scope separation.

```

219d1ec (HEAD -> main, origin/main) docs: separate portal polish from wallboard redesign
508b9eb docs: complete runtime contract and deployment history
728d74f docs: add portal documentation baseline
51c057e docs: define portal documentation roadmap
0d27e67 feat: redesign wallboard and stabilize portal visuals
7f007cf refine: deepen public wallboard visual system
094ba0d feat: add public TV operations wallboard
8ee0b1b feat: expose manager crew role and framework weights
7a00d5d fix: sync billing adapter with live column schema
9d477b2 feat: add Customer Repository billing portal adapter
88accf2 fix: finalize billing filters and source states
73d3f4d docs: capture Monday workflow safeguards
79f26ca fix: make analytics network initialization reliable
1ca6d4b fix: keep analytics panels on one desktop row
f7a5366 feat: add read-only Monday workflow network adapter
a8ae0c9 restore: previous final zoom out animation
31afe6e fix: preserve camera position in final wide shot
60816d5 fix: smooth cinematic workflow playback

```

Current uncommitted changes at handoff: M .gitignore, M README.md, M docs/data-contract.md, M docs/monday-source-map.md, M scripts/portal_server.py, M scripts/sync_crew_repository.py, M scripts/validate_crew_snapshot.py, M tests/test_crew_repository.py, ?? docs/SkyStation-Operations-HQ-Project-Handoff.pdf, ?? scripts/sync_role_framework.py, ?? tests/test_role_framework.py. These changes were deliberately not included in the documentation commits and must be reviewed before any cleanup, reset, or merge.

Do not reset or clean the working tree casually. The uncommitted changes may represent ongoing source/role-framework work. Start by reviewing git diff and the untracked files with the owner.

12. What is complete, partial, and next

State	Items
Complete	Portal preview shell; public wallboard static release; six-screen cycle; favicon/back link; Analytics first-load fix; source contracts; adapter/test foundations; documentation baseline.
Partial	Live public projection; Manager Vault; production API deployment; role framework integration; complete public data wiring; polished visual QA; route-level auth UX.
Not started / external	Production HTTPS hosting; secret manager; scheduler; Google OAuth production config; manager allowlist; snapshot retention policy; canonical logic Doc URL registration.
Next engineering step	Review and commit the uncommitted role-framework/source changes safely, then wire validated projections into the runtime before claiming live data.
Next design step	Internal portal: follow polish-only audit on a branch. Public Dashboard: plan a broader redesign separately.
Next verification step	Restore a real browser/screenshot path and run visual QA at 1920×1080, 1366×768, and narrow fallback.

13. Handoff procedure for a new owner

- Clone/open the repository and read this PDF first, then docs/documentation-roadmap.md.
- Run `git status --short` and preserve the uncommitted source/role-framework changes until the owner confirms their disposition.
- Read `PRODUCT.md`, `docs/product-requirements.md`, `docs/business-logic-index.md`, and `docs/data-contract.md` before changing data or metrics.
- Read `docs/architecture.md`, `docs/api-contract.md`, and `docs/security-and-privacy.md` before adding or exposing an endpoint.
- Run the test baseline with `uv run --with pytest pytest -q`.
- Inspect `rebuild-preview.html` and `public-wallboard.html` as separate products. Do not apply internal portal polish constraints to the future wallboard redesign.
- For any feature, update the relevant product/design/data/architecture/security/ADR/QA/runbook documents in the same slice.
- Before release, verify source, tests, public scan, browser rendering, deployment, and rollback evidence. Do not claim a check that did not run.
- Never persist secrets; replace any discovered values with `[REDACTED]`.

End state for this handoff: The project has a substantial working preview and an explicit path to production, but it is not yet a production-connected, browser-visual-QA-approved operations portal. The next owner should preserve the data and privacy boundaries while finishing runtime integration and separately refining the two design tracks.